

Permutation and combination

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1 Definitions

Permutation $\longrightarrow$ arrangements $\longrightarrow {}^nP_r$ permutation

Combination $\longrightarrow$ selection $\longrightarrow {}^nC_r$ combination

2 Q1 — Words from DELHI

(Q.1) No of words formed by word “DELHI” (positions 1 2 3 4 5)

without repetition

5	4	3	2	1
---	---	---	---	---

$$= 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

with repetition :-

5	5	5	5	5
---	---	---	---	---

$$\text{No of ways} = 5 \times 5 \times 5 \times 5 \times 5 = 5^5$$

3 Q2 — Words from ALLAHABAD

(Q.2) Do same if word is “ALLAHABAD” (positions 1 2 3 4 5 6 7 8 9)

$$\text{No of arrangements} = \frac{9!}{4!2!}$$

(the 4! is the *no of times A repeated*; the 2! is the *no of times 'L' is repeated*)

4 Q3 — Words from AALLBBCOEFGG

(Q.3) Do same if word is “AALLBBCOEFGG” (positions 1 2 3 4 5 6 7 8 9 10 11 12)

Assume all letters are different = 12!

But A is repeated 2 times (No of ways of arranging 2) = 2!

L $\rightarrow$ repeated 2 $\rightarrow$ = 2!

B $\rightarrow$ repeated 2 $\rightarrow$ = 2!

G $\rightarrow$ repeated 2 $\rightarrow$ = 2!

So Correct no of ways

$$= \frac{12!}{2!2!2!2!}$$

5 Q4 — LUCKNOW, vowels always together

(4) How many words can be formed by using all the letters of the word “LUCKNOW” so that the vowels always come together?

(separating vowels & assuming these a single letter)

“LUCKNOW” $\uparrow \uparrow$ (the vowels are U and O)

<i>OU</i>	<i>L</i>	<i>C</i>	<i>K</i>	<i>N</i>	<i>W</i>
1	2	3	4	5	6

No of ways of arranging 6 obj. = $6! \times 2!$

($2!$ is the No of ways of arranging the 2 objects, i.e. the 2 vowels inside their block)

Total arrangements = $6! \times 2!$

6 Q5 — SUCCESS, vowels always together

(5) Do same is word is “SUCCESS”

SUCCESS (vowels underlined: U, E)

<i>UE</i>	<i>S</i>	<i>C</i>	<i>C</i>	<i>S</i>	<i>S</i>
1	2	3	4	5	6

$$\text{No of ways} = \left(\frac{6!}{2!3!} \right) \times 2!$$

(‘C’ is repeated 2 times; ‘S’ is repeated 3 times; the $\times 2!$ arranges the vowels — No of ways of arranging vowels)

7 Q6 — LUCKNOW, vowels never together

(6) How many words can be formed by using all the letters of the word “LUCKNOW” so that the vowels never come together?

“LUCKNOW” (positions 1 2 3 4 5 6 7)

No of ways vowels never come together

$$= \text{Total arrangements} - \text{Arrangements in which vowels come together}$$

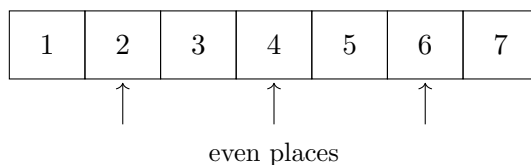
$$= 7! - 6! \times 2!$$

$$= 7 \times 6 \times 120 - 6 \times 120 \times 2 = 5 \times 6 \times 120$$

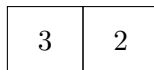
$$= 3600$$

8 Q7 — LUCKNOW, vowels occupy even places

(7) How many words can be formed by using all the letters of the word “LUCKNOW” so that the vowels occupy even places?



Arranging 3 places for 2 vowels.



$$= 3 \times 2 = {}^3P_2 = 6 \text{ ways.}$$

(No of places available for vowel 1; No of places available for vowel 2)

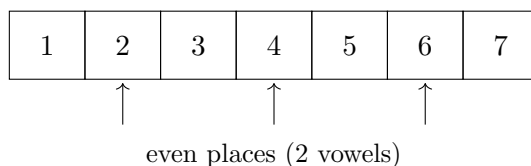
Once vowels have been placed we have 5 places left for 5 letters (LCKNW).

$$\text{No of ways} = 6 \times 5! = 6!$$

9 Q8 — SUCCESS, vowels occupy even places

(8) How many words can be formed by using all the letters of the word “SUCCESS” so that the vowels occupy even places?

SUCCESS (vowels: U, E)



= No of ways vowels can occupy even places $\times$ Total no of ways of arranging remaining letters

$$\begin{aligned}
 &= {}^3P_2 \times \frac{5!}{3!2!} \\
 &= \frac{3!}{1!} \times \frac{5!}{3!2!} = 5 \times 4 \times 3 = \underline{60}
 \end{aligned}$$

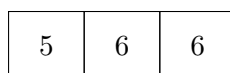
10 Q9 — Numbers between 99 and 1000

Ques: How many numbers between 99 and 1000 can be formed with the digits 0, 1, 2, 3, 4, 5

We need to form 3-digit numbers.

Total no of objects to be arranged = 0, 1, 2, 3, 4, 5 = 6 (we can not have ‘0’ [leading]; can include zero elsewhere)

with repetition $\rightarrow$



$$= 5 \times 6 \times 6 = 180 \text{ ways}$$

without repetition $\rightarrow$

5	5	4
---	---	---

$$= 5 \times 5 \times 4 = 100 \text{ ways.}$$

Note :- If not asked “with repetition / without repetition” & both answers are ready, then select the answer in which repetition is allowed.

11 Q10 — 6-digit numbers from 2,3,4,5,6,8

(10) How many 6 digit numbers can be formed by the digits 2, 3, 4, 5, 6, 8 when

(i) All the no’s are divisible by 2

with repetition

6	6	6	6	6	4
---	---	---	---	---	---

$$= 6^5 \cdot (4) \quad (4 = \text{no of even digits})$$

without repetition →

1	2	3	4	5	4
---	---	---	---	---	---

$$= 4 \times 5! = 480 \text{ ways}$$

(ii) Every no. is greater than 400000

with repetition

4	6	6	6	6	6
---	---	---	---	---	---

$$= 4 \times 6^5 \quad (\text{no of digits} \geq 4)$$

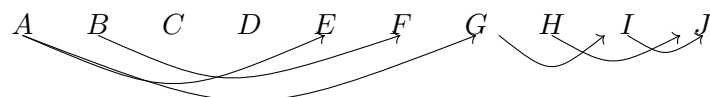
without repetition

4	5	4	3	2	1
---	---	---	---	---	---

$$= 4 \times 5! = 480 \text{ ways.}$$

12 Q11 — Handshakes among 10 people

(11) Ques: There are 10 people in a party. If each one shakes hand with the other exactly once, find the no. of handshakes?



A will shake hands with 9 people

B $\longrightarrow$ 8 people

C $\longrightarrow$ 7

D $\longrightarrow$ 6

(and so on down to 1)

$$\text{Total} = 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = \frac{9 \times 10}{2} = 45 \text{ handshakes.}$$

or :- No of ways of selecting 2 people for handshake among 10.

$${}^nC_2 = {}^{10}C_2 =$$

10	9
----	---

$$= \frac{9 \times 10}{2!} = 45$$

(No of ways of arranging 2 people)

13 Q12 — Handshakes = 66, find no of people

(12) If total no. of handshaking is 66, find the no. of people?

(a) 10 (b) 11 (c) 12 (d) 13

$${}^nC_2 = 66 \Rightarrow \frac{n!}{(n-2)!2!} = 66$$

$$\frac{n(n-1)}{2} = 66$$

$$n(n-1) = 132$$

$$n^2 - n - 132 = 0$$

$$n = \frac{1 \pm \sqrt{1+528}}{2} = \frac{1 \pm \sqrt{529}}{2} = \frac{1+23}{2}$$

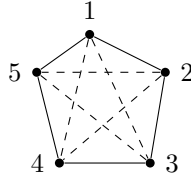
$$(n = 12)$$

14 Q13 — Diagonals of a 10-point convex polygon

(13) If there are 10 points, find the no of diagonals?



for ($n = 4$)



diagonals = 5 : (1-3, 1-4), (2-4, 2-5), (3-5)

[A photo — a hand pointing at a whiteboard worked example — was pasted into the notebook at this point. It re-derives the $n = 4$ case: ${}^4C_2 = \frac{4 \times 3}{2} = 6$, diagonals = $6 - 4 = 2$, boxed alongside ${}^nC_2 - n = \frac{n(n-3)}{2}$.]

Points 1–10: listing all pairs and striking out sides (adjacent pairs) and repeats —

(1-2)[×], 1-3, 1-4, 1-5, 1-6, 1-7, 1-8, 1-9, (1-10)[×]

(2-3)[×], 2-4, 2-5, 2-6, 2-7, 2-8, 2-9, 2-10, (2-1)[×]

(3-4)[×], 3-5, 3-6, 3-7, 3-8, 3-9, 3-10, (3-1)[×], (3-2)[×]

... (and so on for points 4 through 10, each time crossing out the two adjacent sides and any pair already listed)

Using combinations, we need to choose two points from 10 points.

10	7
----	---

$$= \frac{70}{2!} \quad \Leftarrow \text{No of arrangements of 2 obj.}$$

$$= 35$$

[As written: the sentence above the box announces “choose two points from 10” (${}^{10}C_2 = 45$), but the box itself already applies the diagonal shortcut $n(n-3) = 10 \times 7 = 70$, giving $70/2 = 35$ diagonals directly — not ${}^{10}C_2$. The final value (35) is correct for the decagon; the two lines just use different formulas.]

For ‘n’ points we have

$$\left. \frac{n \times (n-3)}{2} \right|_{n=4} = \frac{4 \times 1}{2} = 2 \checkmark$$

$$\text{for } (n=5) : \text{no of diagonals} = \frac{5 \times 2}{2} = 5 \checkmark$$

$$\text{no. of diagonals} = \frac{n(n-3)}{2} \quad (\text{Domain : } n-3 \geq 0, n \geq 3)$$

for $n = 3$: diagonals = 0

OR

$${}^nC_2 - n = \frac{n!}{(n-2)!2!} - n = \frac{n(n-1)}{2} - n = \frac{n^2 - n - 2n}{2} = \frac{n^2 - 3n}{2} = \frac{n(n-3)}{2}$$

Ques: If there are 10 points, find the no. of diagonals?

Solⁿ:- 1 line $\rightarrow {}^nC_2 \quad \triangle \rightarrow {}^nC_3 \quad \text{For ‘n’ points :-}$

No of straight lines = nC_2

No of Triangles = nC_3

No of quadrilateral = nC_4

nC_2 = Total no. of straight lines (including diagonal)

Normal straight lines = n (So No of diagonals = ${}^nC_2 - n$)

15 Q14 — Team of 11 from 17 players with 2 bowlers

(14) Out of 17 players, there are 5 bowlers. Find no of ways of selecting a team of 11 players which has 2 bowlers.

No of ways of selecting 2 bowlers from 5 = 5C_2

No of ways of selecting 9 batsman out of 12 = ${}^{12}C_9$

$$\begin{aligned}\text{Total No of ways} &= {}^5C_2 \times {}^{12}C_9 = \frac{5!}{3!2!} \times \frac{12!}{9!3!} \\ &= \frac{5 \times 4}{2} \times \frac{12 \times 11 \times 10}{3 \times 2} \\ &= 20 \times 110 = \underline{2200}\end{aligned}$$

16 Q15 — Committee of 3 from 5 males and 4 females

(15) Ques: A committee of 3 members is to be formed out of 5 males and 4 females. Find how many committees can be formed consisting of -

(i) 2 Male and 1 female in the committee

(ii) No female

(iii) At least 1 female

(iv) At most 1 female

(i) No of ways of selecting 2 male out of 5 males = 5C_2

No of ways of selecting 1 female out of 4 = 4C_1

So Total no of ways of selecting committee

$$\begin{aligned}&= {}^5C_2 \times {}^4C_1 = \frac{5!}{2!3!} \times \frac{4!}{1!3!} \\ &= 4 \times \frac{5 \times 4}{2} = 10 \times 4 = \underline{40}\end{aligned}$$

(ii) No of ways of selecting 3 males out of 5 = ${}^5C_3 = 5 \times 4/2 = 10$

0 females $4 = {}^4C_0 = 1$

Total no. of ways = 10

(iii) (F, M) $F + M = 3$

$(1, 2) + (2, 1) + (3, 0)$

$${}^4C_1 \times {}^5C_2 + {}^4C_2 \times {}^5C_1 + {}^4C_3 \times {}^5C_0 = 4 \times \frac{5 \times 4}{2} + \frac{4 \times 3}{2} \times 5 + 4 \times 1$$

$$= 40 + 30 + 4 = \underline{74}$$

(OR) No of ways of at least one Female

$$= \text{Total} - \text{None female}$$

$$= {}^{N_m+N_F}C_3 - {}^{N_m}C_3$$

$$= {}^{5+4}C_3 - {}^5C_3 = {}^9C_3 - {}^5C_3 = \frac{9 \times 8 \times 7}{3 \times 2} - \frac{5 \times 4}{2} = 84 - 10 = \underline{74}$$

(iv) $N(\text{At most 1 female}) = ?$ $5M, 4F$ ($F + M = 3$)

$$(M, F) = (3, 0) + (2, 1)$$

$${}^5C_3 \times {}^4C_0 + {}^5C_2 \times {}^4C_1 = \frac{5 \times 4}{2} \times 1 + \frac{5 \times 4}{2} \times 4$$

$$= 10(1 + 4) = \underline{50}$$

17 Q16 — Vegetables MCQ

(16) Ques: How many different vegetables can be made from Cauliflower, tomatoes, onions, potatoes and carrots?

(a) 16 (b) 28 (c) 31 ✓ (d) 32

$$N = {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5$$

$$= 5 + \frac{5 \times 4}{2} + \frac{5 \times 4 \times 3}{3 \times 2} + 5 + 1$$

$$= 5 + 10 + 10 + 5 + 1$$

$$= 30 + 1 = \underline{31}$$

18 Q17 — Handshakes in a group of 11 MCQ

2018-June-Physics:

(17) Ques: In a group of 11 persons, each shakes hand with every other once and only once. What is the total number of such handshakes?

(a) 110 (b) 121 (c) 55 ✓ (d) 66

No of hand shakes = No of ways of selecting 2 people

$$= {}^{11}C_2 = \frac{11 \times 10}{2} = \underline{55}$$

19 Q18 — 3-letter words, consonant condition MCQ

Dec-2017 (Physics):

(18) Ques: The number of 3 english letter words, having at least one consonant but not having two consecutive consonants is:

- (a) 2205 (b) 3780 ✓ (c) 2730 (d) 3360

No of letters = 26

No of consonants = $26 - 5 = 21$

$$N = \text{Total ways} - N(\text{none consonant}) - N(2 \text{ consecutive consonant})$$

$$= 26 \times 26 \times 26 - 5 \times 5 \times 5 - \dots$$

[This route is crossed out ("OR") in favour of the case-based method below.]

(C: consonant)

(m) V V C

(n) V C V

C V V } $\rightarrow N_1$

(m) 2C: C V C $\rightarrow N_2$

(or) 3C: C C C — 2 consecutive consonants, so this case won't contribute $\rightarrow N_3 = 0$

(Required) $N_T = N_1 + N_2 + \cancel{N_3} = ?$ (Total ways)

5	5	21
5	21	5
21	5	5

$$N_1 = 5 \times 5 \times 21 \times 3 = 75 \times 21 = 25 \times (20 + 1) = 1500 + 75 = 1575$$

C	V	C
21	5	21

$$N_2 = 21 \times 21 \times 5 = 441 \times 5 = 2205$$

Required Ways = $N_1 + N_2 = 1575 + 2205 = \underline{3780}$

20 Q19 — Diagonals of a dodecagon MCQ

Dec-2015 (Chemistry):

(19) Ques: The number of diagonals of a convex dodecagon (12-gon) is:

- (a) 66 (b) 54 ✓ (c) 55 (d) 60

No of points: $n = 12$

$$\text{No of diagonals} = \text{Total lines} - \text{normal lines}$$

$$= {}^nC_2 - n$$

$$= \frac{n(n-3)}{2} = \frac{12 \times 9}{2} = \underline{54}$$

21 Q20 — Letters among 20 friends MCQ

JUNE-2013:

(20) Ques: During a summer vacation, of 20 friends from a hostel, each wrote a letter to each of all others. The total number of letters written was -

- (a) 20 (b) 400 (c) 200 (d) 380 ✓

Solⁿ:-

Any one will write 19 letters

So Total letters = 19×20

$$= \underline{380}$$

[Unlike the handshake problems, writing a letter isn't mutual, so this uses 19×20 directly rather than ${}^{20}C_2$.]

22 Q21 — Arrangements of UGCCSIR MCQ

(21) Qs: What is the number of distinct arrangements of the letters of the word “UGCCSIR” so that U and I can not together?

- (a) 2520 (b) 720 (c) 1520 (d) 1800 ✓

$$N(\text{'UI' can't come together}) = N_T - N(\text{UI come together}).$$

N_T = total no of words = “U G C C S I R” (positions 1 2 3 4 5 6 7)

$$= \frac{7!}{2!} = 7 \times 6 \times 5 \times 4 \times 3 = 7 \times 360 = 2520$$

No of ways in which UI come together

$$= (UI) GCCSR \quad (\text{positions } 1-2-3-4-5-6)$$

$$= \frac{6!}{2!} \times 2! = 6! = 6 \times 120 = 720$$

$$N_{req} = N_T - 6! = 2520 - 720 = \underline{1800}$$

23 Revisited — Numbers between 99 and 1000 again

Ques: How many numbers between 99 and 1000 can be formed with the digits 0, 1, 2, 3, 4 & 5

[As written: ‘0,’ and ‘1,’ are struck through with an X in the notebook, but the working below still draws from a 6-digit pool — matching Q9’s digit set $\{0, 1, 2, 3, 4, 5\}$ exactly, not the reduced $\{2, 3, 4, 5\}$ the strike-through would suggest.]

Solⁿ:- 99 – 1000

(without repetition) rep. allowed

$$5 \times 5 \times 4 = 100 \text{ Ans.}$$

$$5 \times 6 \times 6 = 180 \text{ Ans.}$$